

10-606 Recitation

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Warmup

True or False: Let $\mathbf{X} \in \mathbb{R}^{n \times m}$, $\mathbf{Y} \in \mathbb{R}^{n \times k}$, and $\boldsymbol{\theta} \in \mathbb{R}^{m \times k}$. Then the following equality holds (recall that the trace of a matrix, tr , is the sum of its diagonal elements):

$$\frac{1}{2} \text{tr}((\mathbf{X}\boldsymbol{\theta} - \mathbf{Y})^T(\mathbf{X}\boldsymbol{\theta} - \mathbf{Y})) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^k (\mathbf{X}\boldsymbol{\theta} - \mathbf{Y})_{ij}^2$$

Solution: $\text{tr}(\mathbf{A}^T \mathbf{A}) = \sum_{i=1}^n \sum_{j=1}^k (\mathbf{A})_{ij}^2$ where $\mathbf{A} \in \mathbb{R}^{n \times k}$. Then,

$$\text{tr}(\mathbf{A}^T \mathbf{A}) = \sum_{j=1}^k (\mathbf{A}^T \mathbf{A})_{jj} = \sum_{j=1}^k \sum_{i=1}^n (\mathbf{A})_{ij}^2 \text{ because } (\mathbf{A}^T \mathbf{A})_{jj} = \sum_{i=1}^n \mathbf{A}_{ij}^2.$$

Linear Operators, Rank, Eigenvalues

Definitions/Basic Properties

1. Define a *functional* and give an example.

Solution: A functional is a type of function whose input is a function and whose output is a scalar, with a mapping such as $V \rightarrow \mathbb{R}$. Example: $f(x) = \langle v, x \rangle$ for some v .

2. What does it mean for a function $f : V \rightarrow V$ to be *linear*?

Solution: $f(ax + by) = af(x) + bf(y)$ for all scalars a, b and vectors $x, y \in V$.

3. Prove that a linear operator maps the origin to the origin. That is, if f is linear, then $f(0) = 0$.

Solution: Since $0 + 0 = 0$, we have

$$f(0 + 0) = f(0).$$

Since f is linear, the left side is equal to

$$f(0) + f(0) = 2f(0).$$

Therefore,

$$2f(0) = f(0),$$

and subtracting $f(0)$ from both sides gives

$$f(0) = 0.$$

4. Prove that if A is positive semi-definite then its eigenvalues are nonnegative.

Solution: Suppose A is PSD with eigenvalue λ and corresponding eigenvector v . Then

$$Av = \lambda v.$$

Multiplying each side by $v^\top$ gives

$$0 \leq v^\top Av = v^\top \lambda v = \lambda v^\top v = \lambda \|v\|_2^2.$$

Since $\|v\|_2^2 > 0$ for a nonzero eigenvector, it follows that

$$\lambda \geq 0.$$

Conceptual Understanding

5. What's the geometric meaning of having multiple linearly independent eigenvectors corresponding to the same eigenvalue?

Solution: The transformation stretches vectors by the same amount in all of those directions.

6. What happens if an eigenvalue of a matrix is equal to zero? Interpret this geometrically.

Solution: When an eigenvalue of a matrix equals zero, the linear transformation collapses at least one dimension of the space by mapping nonzero vectors onto the zero vector. The matrix is then not invertible, since the determinant is the product of its eigenvalues, and therefore singular.

Geometrically, it squashes vectors along the eigenvector direction to 0, reducing dimensionality. The number of zero eigenvalues is related to the dimension of the nullspace, and the matrix loses full rank.

Applied Problems

7. Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

- (a) Compute its eigenvalues and eigenvectors.
(b) Is it positive semi-definite?

Solution:

To find the eigenvalues λ , solve the characteristic equation $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 0 & 2 - \lambda \end{pmatrix} = 0.$$

Since the matrix is upper triangular,

$$(2 - \lambda)^2 = 0.$$

Thus,

$$\lambda_1 = 2, \quad \lambda_2 = 2.$$

To find the eigenvectors corresponding to $\lambda = 2$, solve

$$(A - 2I)v = 0 :$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives

$$v_2 = 0.$$

The variable v_1 is free. Setting $v_1 = 1$, we obtain

$$v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Therefore, the eigenvalues are $\lambda = 2$ (with algebraic multiplicity 2), and the eigenvectors are all vectors of the form

$$\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \alpha \in \mathbb{R}.$$

A is not symmetric, so it is not positive semi-definite.

8. Suppose a linear transformation ℓ has rank 3. What can you say about the dimension of its null space if $\ell : \mathbb{R}^5 \rightarrow \mathbb{R}^5$?

Solution: The rank-nullity theorem states

$$\text{rank}(\ell) + \text{nullity}(\ell) = \dim(\text{domain}).$$

Therefore,

$$\dim(\ker \ell) = 5 - 3 = 2.$$

Matrices & Systems of Equations

Definitions/Basic Properties

1. Prove that the system $Ax = b$ has a unique solution if A is full rank.

Solution:

Suppose x_1 and x_2 satisfy the system:

$$Ax_1 = b, \quad Ax_2 = b.$$

Then

$$A(x_1 - x_2) = 0.$$

Because A has full rank, it is invertible. Thus,

$$A^{-1}A(x_1 - x_2) = A^{-1}0,$$

which gives

$$I(x_1 - x_2) = 0.$$

Therefore,

$$x_1 - x_2 = 0,$$

so

$$x_1 = x_2.$$

Hence, the solution is unique.

2. What are the three possible outcomes for a system of linear equations depending on the rank of its coefficient matrix relative to the number of unknowns?

Solution:

- Rank $>$ number of unknowns: impossible, since rank is at most the minimum of the number of rows and columns.
- Rank = number of unknowns: unique solution, assuming the system is consistent.
- Rank $<$ number of unknowns: either infinitely many solutions or no solutions.

Conceptual Understanding

3. Why do elementary row operations preserve the solution set of a linear system?

Solution: They correspond to algebraic manipulations of equations (swapping, scaling, or adding multiples of equations), which do not change the set of solutions. Further, each operation is reversible.

4. Explain why the number of nonzero rows in REF equals the rank of the matrix.

Solution: Performing row operations on a matrix with a linearly dependent row results in a row of all zeros as elementary row operations preserve linear dependence. Therefore, the number of nonzero rows in REF equals the rank of the matrix because every resulting nonzero row is linearly independent from each other.

Applied Problems

5. Solve the system using Gaussian elimination:

$$\begin{cases} x + y + z = 3 \\ 2x + 3y + z = 7 \\ x + 2y + 2z = 5 \end{cases}$$

Solution:

$$\begin{aligned} & \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 7 \\ 1 & 2 & 2 & 5 \end{array} \right] \\ & \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}]{} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 1 & 1 & 2 \end{array} \right] \\ & \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 2 & 1 \end{array} \right]. \end{aligned}$$

Subsequently,

$$x = 1, \quad y = \frac{3}{2}, \quad z = \frac{1}{2}.$$

6. Solve the system using Gaussian elimination:

$$\begin{cases} 2x + y - z = 1 \\ 4x - 6y = -2 \\ -2x + 7y + 2z = 9 \end{cases}$$

Solution:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array} \right]$$

$$\xrightarrow{\begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{matrix}} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & -8 & 2 & -4 \\ 0 & 8 & 1 & 10 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & -8 & 2 & -4 \\ 0 & 0 & 3 & 6 \end{array} \right].$$

Subsequently,

$$x = 1, \quad y = 1, \quad z = 2.$$